Journal of Oral Rehabilitation 2002 **29**; 911–917

Localization of TMJ sounds to side

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SUMMARY **Differential diagnosis depends in cases with disk displacement on accurate identification of sound source. Mistakes may occur when clicking from one temporomandibular joint (TMJ) is heard on both sides of the head at auscultation and neither examiner nor patient, is sure about side. The hypothesis was that the head tissues affect spectral characteristics of TMJ sounds and that differences due to different positioning of sensors can be used in localization of source. The aim was to compare bilateral electronic recordings of unilateral TMJ sounds to obtain and compare attenuation, phase shift and time delay. Recordings were made from 12 subjects with unilateral clicking. Small electret condenser microphones, bandwidth 40–20 000 Hz, were placed at the openings of the auditory canals and the sounds were recorded at a sampling rate of 48 000 Hz. The head tissues acted as a filter causing a frequency dependent attenuation and phase shift. There was a time difference between the ipsi- and the contra lateral recordings, the latter always having a longer delay time (range 0·2–1·2 ms, group mean 0·68 ms, s.d. 0·292 ms). In conclusion, spectral analysis of bilateral electronic TMJ sound recordings is of diagnostic value when bilateral clicking is heard at auscultation and can help to avoid diagnosing a silent joint as clicking.**

KEYWORDS: **clicking, diagnosis, disk displacement, Fourier analysis, signal processing, temporomandibular joint, transfer function**

Introduction

Temporomandibular joint (TMJ) sounds indicate temporomandibular dysfunction (TMD). Accurate localization of the side is important when differing between types of disk displacement. For instance, TMJ clicking during opening and closing is considered to be a sign of disk displacement with reduction (Farrar, 1972; Isberg-Holm & Westesson, 1982; Isberg-Holm, Widmalm & Ivarsson, 1985). Recording of this sign is important in the evaluation of treatment outcome (Le Bell & Kirveskari, 1985). As pointed out by Watt (1966) it is, however, often difficult to decide from which side (joint) a clicking comes when it can be heard at auscultation on both sides. That may cause silent joints to be classified as clicking. Sometimes, the patients' own observations can be of help but not always, for instance with small children, with patients with impaired hearing, emotionally unbalanced, mentally handicapped patients, or with patients having speech or language communication problems. Even adults, mentally alert subjects, are not always sure about the side of clicking. Accurate screening methods may be especially important in examination of small children who for some reason cannot make or communicate their own reliable observations because significant signs of temporomandibular disorders occur already in paediatric groups, 4–6 years of age (Widmalm *et al*., 1995; Widmalm, Christiansen & Gunn, 1999a). Internal derangement was a frequent finding in a paediatric group (age 8–16) of patients with pain and dysfunction of the TMJ as reported by Katzberg *et al*. (1985) who emphasized the importance of initiating treatment as soon as possible. Those who acknowledge the importance of early recognition have therefore a need for a more reliable method of source localization to avoid being dependent on the examiner's and/or the patient's subjective opinions. It has been suggested (Widmalm, Williams & Yang, 1999b) that electronic recording of TMJ sounds might be used for localization of sound source by comparing the signal characteristics of sounds recorded bilaterally.

The *hypothesis* was that TMJ sounds travelling through the head tissues are subject to filtering, causing the bilateral recordings of a single clicking to show differences in the time and frequency domains that can be used to decide from which side the sound comes.

The *aim* of the present study was to compare amplitude and phase shift between bilateral recordings of unilateral clicking sounds and to calculate the relative time delay between the recordings.

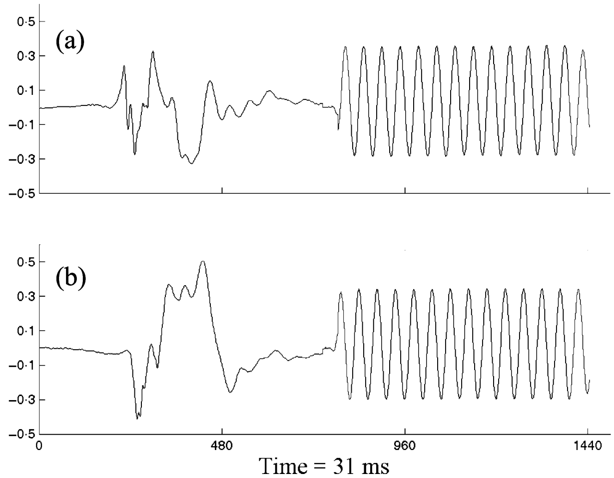

Fig. 1. This figure illustrates the calibration procedure used to check that the gain was the same in both channels. Sampling rate was 48 kHz. The upper graph (a) is from the left side microphone and the lower, (b) is from the right side. Each sound recording is followed by a calibration signal, 94 dB and 1000 Hz. Vertical axes are amplitude in parts of maximal (+1·0) and minimal (−1·0) values. The horizontal axes are in samples. 480 samples = 10 ms. The clicking in (a) has an amplitude of 94 dB and the clicking in (b) is 3 dB higher. It should be noted that the sound dB level is not the same as sound pressure level (SPL) or loudness level. We are comparing the amplitudes between two electrical signals from the microphones. The exact relationship between measurements of the above type and true SPL recordings will be subject to further studies. The method used in this paper is, however, adequate for comparing amplitude levels between bilateral recordings with the aim to localize the true sound source.

Materials and methods

Subjects

The group consisted of 10 females and two males, age range 22–74, with unilateral TMJ clicking that could be heard with similar loudness at both sides at auscultation. We use the term *bilateral clicking* for such clicking when we suspect it comes from one joint only. All subjects were sure that the clickings occurred only on one side and that they could recognize on which side the clickings occurred. The subjects were fully informed of the experiments and agreed to take part.

Electronic recording

Small electret condenser microphones (diameter 5 mm)*, bandwidth 40–20 000 Hz, placed at the openings of the external auditory canals, were used for recording to a 16 bit digital tape recorder†. The sampling rate was 48 000 Hz. A Mackie, Microseries 1202 Mic/Line Mixer‡ with adjustable gain settings was used for amplification of the microphone signals. Amplitude calibration was achieved by recording a 94 dB, 1000 Hz, signal§ (Fig. 1) with each microphone at the start of each session and again if gain settings were changed.

Bilateral TMJ sound recordings were made while the subjects performed wide jaw opening-closing movements at a rate of one cycle per second. They were asked to perform about 15–20 cycles if possible without feeling pain or discomfort. One clicking recording was selected from each cycle for analysis. If clicking occurred both during opening and closing the one with the largest peak-to-peak amplitude was selected.

Statistical methods

The recordings were analysed using Cool Edit Pro¶, Matlab** software and application programs written in Matlab. The transfer function of the clicking sounds was calculated to compare magnitude, phase shift and time delay between the contra- and ipsi-laterally recorded sounds. It was evaluated by comparing the bilateral

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‡Mackie Designs, Woodinville WA, USA.

§Sound Level Calibrator Type 4231, Brüel & Kjær, A/S, Nærum, Denmark.

¶Syntrillium Software Corporation, Phoenix, AZ, USA.

**The MathWorks Inc., Natick, MA, USA.

recordings of a unilateral TMJ clicking sound (Figs 2 and 3). If the ipsi-lateral sound recording clicking is treated as the input, $o(t)$, and the sound recorded on the contra lateral side as the output, $e(t)$, the transfer function is expressed as the Fourier transform of the output divided by the Fourier transform of the input. Most textbooks describe taking the *Laplace transform* of the input/output differential equation and generating a transfer function (Balmer, 1997). In this paper a slightly different technique was used (Williams, Gesink, & Stern, 1972). The transfer function was calculated as follows:

$o(t)$ = the original sound; $e(t)$ = the sound recorded on the contra-lateral side; $O(f)$ = the Fourier transform of $o(t)$; $E(f)$ = the Fourier transform of $e(t)$.

Next, for each sound pair (S)††, the auto and cross-spectral densities $S^k_{oo}(f) = O(f)O^*(f)$ and $S^k_{oe}(f) = E(f)O^*(f)$ are formed, where k indicates the k^{th} term, and then the transfer function for k clicking sounds is defined to be

$$H(f) = \frac{\sum_{k=1}^{N} S^k_{oe}(f)}{\sum_{k=1}^{N} S^k_{oo}(f)}$$

If $H(f)$ is assumed to be complex valued and equal to $A(f) + iB(f)$, then the magnitude of

$$|H(f)| = \sqrt{(A(f)^2 + B(f)^2)}$$

and the phase $H(f) = \tan^{-1} [B(f)/A(f)]$.

The terms *phasor, filter,* and *coherence* are briefly described here. More detailed descriptions can be found in textbooks (Steiglitz, 1996; Balmer, 1997).

Phasor

The rotating vector that represents a sinusoid is just a single fixed complex number raised to progressively higher and higher power. The rotating vector arising from simple harmonic motion can be represented by $e^{i\theta} = \cos\theta + i\sin\theta$ (Euler's formula) where θ is the angle, measured in radians. The rotating vector can be rewritten as $e^{i\omega t}$ (ω = angular frequency in radians, t = time). Such a signal is called a *phasor*. It can be regarded as a solution to the differential equation describing the vibrating sound source. It is complex valued but when using it to describe real sound waves we can consider only the real part.

††A 'sound pair' is a clicking recording stored in a wave file with the sound waveform recorded on the left side in one channel and the sound waveform recorded on the other side in the second channel (Fig. 2).

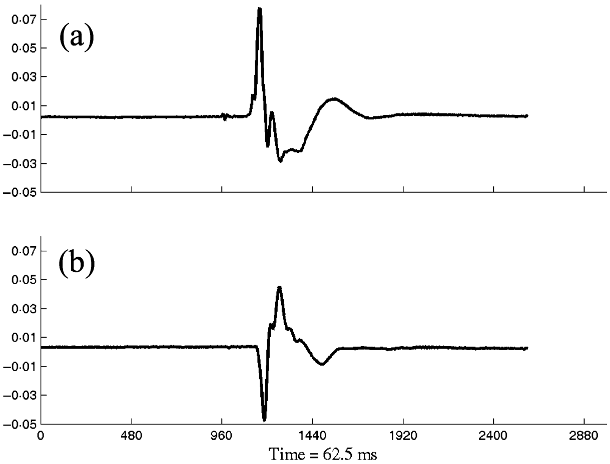

Fig. 2. Recording (subject no. 04) from the left side microphone in the upper channel (a) and from the right side microphone in the lower (b). The numbers of samples are marked on the horizontal axes. 480 samples = 10 ms. The full length of each window is 62·5 ms. Vertical axis is amplitude in parts of maximal range which is +1 to −1 for each window. The recordings have to be relatively 'noise-free'. Otherwise it is not possible to identify the starting point of the sound waveforms. Here the waveform in channel (b) seems to come later in time and may be a delayed recording of the clicking sound in (a). This means that the right TMJ may be silent in spite of a clicking being heard at auscultation also on the right side. The differences in shape are due to the filtering effect of the head tissues. The first deviations are in opposite direction as a result of differences in phase shift.

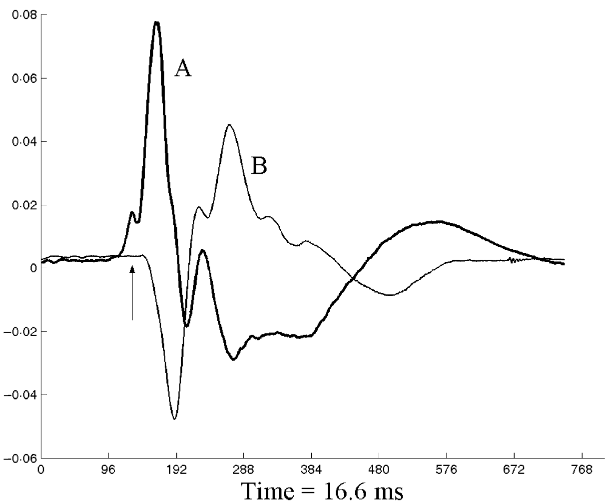

Fig. 3. Close up of the clicking recordings in Fig. 2. (a) (thick line) is from the left side and (b) (thin line) is from the right side. Superimposing makes it easier to observe the delay (arrow). The numbers of samples are marked on the horizontal axis. 96 samples = 2 ms. The full length of the window is 16·6 ms. Vertical axis is amplitude in parts of maximal range which is +1 to −1. The difference in phase shift has caused reverse polarity in the waveform recorded from the right side.

Filter

The spectral characteristics of TMJ sounds are changed because of the filtering effect of the head tissues, presumably with more change the longer distance they have to travel through such tissues before reaching the recording device. Filters work by combining delayed versions of signals. The input phasor (the original TMJ sound) is multiplied by $e^{-i\omega\tau}$ (ω = angular frequency, τ = time delay) which tells us that the filter multiplies the size of the input phasor by the filter's magnitude response, and shifts its phase by the filter's phase response. If the phasor is delayed by τ s we get $e^{i\omega(t-\tau)} = e^{i\omega t}\, e^{i\omega\tau}$ That is a delay of τ s multiplies the phasor by the complex factor $e^{-i\omega\tau}$, that does not depend on time, only on the amount of delay τ and the angular frequency, ω. The original phasor is rotated by the angle $-\omega\tau$, and the magnitude is changed. The slope or rate of change of $\omega\tau$ (derivative of $-\omega\tau$) is the time delay, τ. It is important to understand the effect of this operation on the sound signals because filters combine delayed versions of signals and those are made up of *phasors*.

Coherence

Coherence may be viewed as an indicator of measurement quality. Coherence squared is defined to be $\gamma^2(f) = |S_{oe}(f)|^2/|S_{oo}(f)S_{ee}(f)|$(Williams *et al.*, 1972; Zaveri *et al.*, 1999)

S = sound pair. S_{oo} = auto spectral density of original sound.$S_{oo}(f) = \frac{1}{N}\sum_{k=1}^{N} S_{oo}^{k}(f)$

S_{ee} = auto spectral density of contra laterally recorded sound. $S_{ee}(f) = \frac{1}{N}\sum_{k=1}^{N} S_{ee}^{k}(f)$

S_{oe} = cross-spectral density of original and contra laterally recorded sounds. $S_{oe}(f) = \frac{1}{N}\sum_{k=1}^{N} S_{oe}^{k}(f)$

Coherence ranges between zero and one (or 100%). Zero coherence indicates that the output is not related to the input (the excitation). For a given amount of averaging, the variance of the transfer function will be higher at frequencies where the coherence is low, and lower the closer the coherence is to 100%. In this study a coherence of ≥80% was considered high (Zaveri *et al.*, 1999).

Results

Unilateral TMJ clickings were delayed about 0·7 ms longer in the contra-lateral than in the ipsi lateral recordings. The relative delay was consistent with the negative slope of the phase versus frequency plots (Fig. 4, middle graph). The delay values ranged from 0·2 to 1·2 ms. The group mean (n = 12) was 0·68 ms with a standard deviation of 0·292 ms. Details are given in Table 1.

A frequency dependent phase shift was caused in the sound signals (Fig. 4, middle graph). There were negative phase shifts up to about 12 radians, larger for high compared with low frequencies. Thus higher frequencies were delayed more than the lower.

The head tissues acted on the TMJ sounds as a filter with large variations in magnitude response as a function of frequency (Fig. 4, upper graph). The amplitude differences were often negligible with small frequency differences between the energy peaks. The attenuation effect on different frequency areas was variable, but the pattern was consistent within each recording session (Fig. 5).

Discussion

When TMJ sounds travel through the head tissues their amplitude and phase are affected. With one microphone at the ear canal opening close to the clicking joint and the other microphone at the ear canal on the opposite side there was a relative time delay between the two recordings of the same clicking. This delay is dependent on the distance the sound has to travel but also on the phase shift caused by the filtering effect of the head tissues. It is reasonable to assume that the filtering effect is larger on the sounds that travel through the head to the opposite side. The amount of tissues between the same side of the microphone and the clicking TMJ is small and may be negligible. Whether or not that is true will not affect the results in this study because we were comparing the effects on the contra- and ipsi lateral recordings. There was consistently a relative time delay with the contra-lateral sound recordings always later (0·2–1·2 ms) than the ipsi-lateral recordings. This is of diagnostic value because if clicking sounds are recorded bilaterally and appear consistently with that small relative time delay it can be assumed that the clicking recording appearing later is an artifact, that is, only one joint is clicking.

The total delay is dependent, not only on the velocity of the sound and the distance it travels, but also on the phase shift caused by the tissues the sounds pass through. The total time of the delay can be observed by comparing the analogue recordings but evaluating

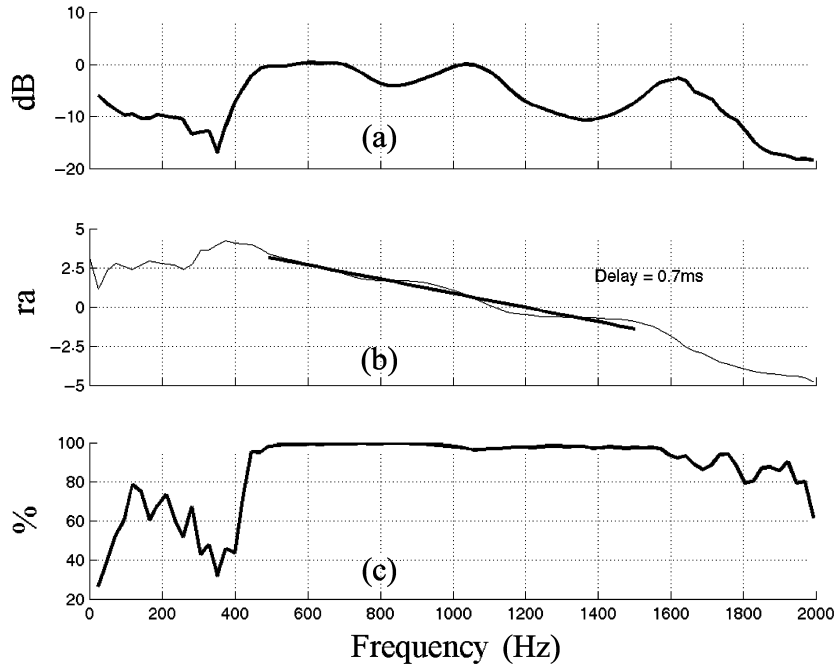

Fig. 4. This figure illustrates how the amplitude and phase were compared between the ipsi- and contra lateral recordings of a unilateral clicking sound from one subject by calculating the transfer function of 13 clicking sounds. Horizontal axes are frequency in Hz. Vertical axes are in decibel (dB) in the upper graph (a), radians (ra) in the middle graph (b), and in percentage in the lower graph (c). The *magnitude response* (a) is displayed as a function of frequency showing the effect of the filter on a phasor of any particular frequency. The dips in the curve mean that the magnitude of the output on the contra lateral side was weaker than the input on the ipsi lateral side, that is the sound intensity was attenuated more in the contra lateral recording but only about 10 dB. Note no differences in attenuation in the frequency area 500–750 Hz and at 1000 Hz. The phase response (b) is displayed as a function of frequency. The negative slope of the graph is consistent with a relative time delay of 0·7 ms. The degree of *coherence* (c) is an indicator of measurement quality that is higher the closer to 100% the coherence is.

Table 1. Time differences (relative delays) between the original unilateral clicking and the contra laterally recorded ‘echo’

Subject ID no.	*n*	Sex	Source	Relative delay in ms
01	5	F	RTMJ	0·3
02	8	F	RTMJ	0·5
03	5	F	LTMJ	0·5
04	13	F	RTMJ	0·7
05	7	F	LTMJ	0·6
06	7	F	LTMJ	0·8
07	7	F	LTMJ	0·2
08	4	F	RTMJ	0·7
09	21	F	LTMJ	0·8
10	11	F	RTMJ	1·1
11	6	M	RTMJ	1·2
12	14	M	RTMJ	0·8

The age range among the 10 female (F) and two male (M) subjects was 22–74. The delay values ranged from 0·2 to 1·2 ms. The group mean was 0·68 ms with a s.d. of 0·292 ms. n = number of clicking recordings, RTMJ = right temporomandibular joint (TMJ), LTMJ = left (TMJ).

the influence of the phase shift needs the use of transfer function calculations.

Localization of TMJ clicking in a side is a diagnostic problem when clicking from one joint is heard with about the same loudness at auscultation at both sides of the head. The results of this study strongly indicate that electronic recording should be used in such cases for more accurate localization by evaluating differences in the time and frequency domains between bilateral recordings of the TMJ sounds. Such information is of value in differential diagnosis of disk displacement. Reciprocal clicking, that is, clicking both during opening and closing, indicates *disk displacement with reduction* (DDR). Not knowing for sure from which joint a clicking comes makes it impossible to know if the clicking really is reciprocal. There is no general agreement about the diagnostic value of reciprocal clicking (Farrar, 1972; Miller *et al.*, 1985; Carlsson, 1999). The results of this study indicate that erroneous

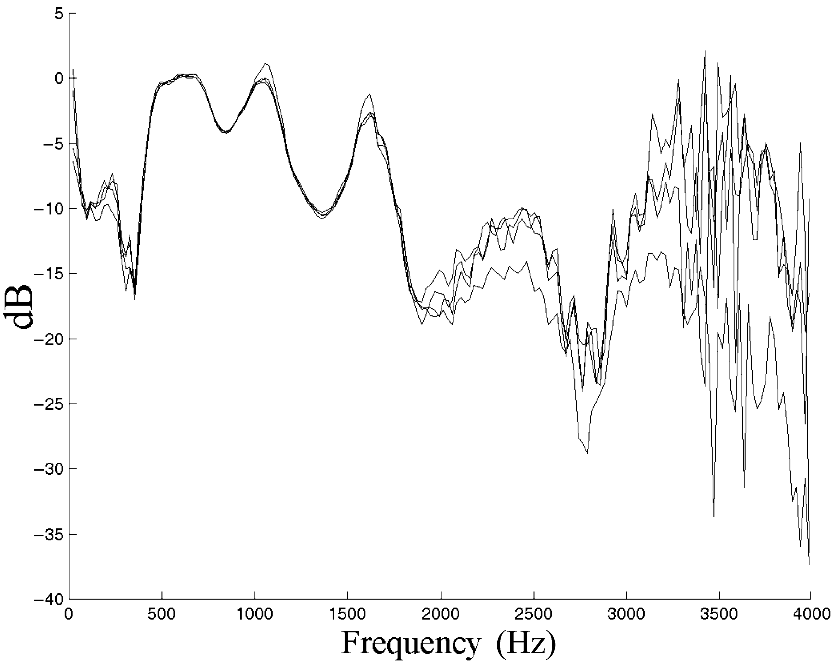

Fig. 5. Thirteen magnitude responses from the same subject as in Fig. 4 are superimposed. Note that the frequency area is extended above 2000 Hz where the variation is very large. This is because the sounds contained very little energy in that frequency area and there the curves only reflect the variation between noise recordings.

classification of silent joints as clicking may have caused part of the confusion.

Bilateral recording and measuring time differences is thus of diagnostic value. The time differences may, however, be as small as 0·2 ms or even less. Sampling rates as low as 600–2000 Hz are not of any use for the type of analysis described here because loud clickings with short duration (1–2 ms) are not always visible in such recordings (Widmalm, Williams & Adams, 1996). If caught at all their waveforms are severely distorted. Sound recording equipment with sampling rates well above 48 kHz may be necessary and will be tried in future studies.

The results of sound velocity measurements in the human skull, as measured on living subjects, vary from 80 to 540 m/s (Zwislocki, 1953; Franke, 1956; Tonndorf & Jahn, 1981). With the microphones placed as in this study the distance from the clicking joint to the contra lateral microphone is about 7 cm longer than to the same-side microphone. The time differences between the bilateral recordings were in the range of 0·2–1·2 ms. The results of our study therefore support that the velocity is in the range of 80–350 m/s. It should be noted that the high velocity figures found in some references, >2000 m/s, are from measurements on dry skulls (Tonndorf & Jahn, 1981) and are not applicable here.

The difference in distance from the sound source to the microphone depends very much upon where the sound originates. The time difference between bilateral recordings of the same TMJ sound may be as small as 0·1 ms or less if the sound 'starts' deep in the glenoid fossa close to the centre of the skull base. The delays can be expected to be larger because of phase shift and differ depending on what path the sound takes: through the mandible, the cranial vault, or the skull base. These factors will be subject to further studies.

In this study, we could only include a minority of our patients, namely those who were absolutely sure about the clicking side. The frequency response may differ between subjects as a result of age, gender, race, size of head and degree of TMJ pathology/dysfunction, etc. These variables could not be considered in this study with only 12 subjects. Further studies on larger groups are needed to evaluate the effect of such factors. The results from our small sample still provide a basic knowledge about the filtering effect of the head tissues. They caused a phase shift that was frequency dependent. Higher frequencies were as expected more delayed than the lower.

The degree of coherence was used an indicator of measurement quality. Coherence analysis has been used in the analysis of human electroencephalograms to study propagation delay. It is a well-developed measure and coherence estimators are designed, in part, to minimize bias and variance (Zaveri *et al*., 1999). It is beyond the scope of this paper to discuss this method in detail and the reader is referred to the literature (Williams *et al*., 1972; Nuttall & Carter, 1976; Zaveri *et al*., 1999).

It was noted that clickings often had amplitudes well above 90 dB (Fig. 1) when comparing with the calibration signal. This does, however, not necessarily mean that the sound pressure level (SPL) was at such high levels. Nor does it give enough information for estimating loudness that is frequency dependent (Kinsler *et al*., 1982).

In conclusion the head tissues had a significant frequency dependent effect on the TMJ sound signals' phase causing part of the total delay. This was always longer, about 1 ms, in the contra- than in the ipsilateral recording. Clinically important is that such differences are too small to be detected by auscultation. Electronic recordings that permit analysis of the TMJ sound signal differences in the time and frequency domains to measure relative time delay are therefore of diagnostic value especially in cases with suspected DDR and when the patient and the examiner are unsure about the side from which the TMJ sound comes.

Acknowledgments

The authors thank Drs Elisabeth Hultcrantz, Stig Arlinger and Alfred Nuttall for reading the manuscript and giving valuable advice.

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